

# Group Comparisons with Analysis of Variance

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## Chapter overview

This chapter compares means by examining variation. One invented study provides the observations throughout: we first compare three groups, then use the full two-factor layout to ask how a comparison changes across practice schedules. We build the one-way ANOVA calculation from individual scores, distinguish its omnibus question from a particular contrast, and explain how the comparison family determines an interval's coverage. Cell means and simple effects then reveal what main effects can conceal. By the end, you will be able to reproduce an ANOVA table, interpret a contrast and its uncertainty, and connect a factorial pattern to the design and model conditions that support its interpretation.

Twenty-four students study the same fractions material and take a common 20-item quiz. In this **invented study**, random allocation places four different students in each combination of three feedback conditions and two practice schedules. *Rereading* offers the worked material again; *Answer feedback* supplies the correct answer after an attempt; *Explanatory feedback* supplies the answer with a worked explanation. Massed practice occurs in one session, and Spaced practice in two shorter sessions with the same total practice time. Everyone takes the quiz at the same specified follow-up point after their final practice. All design details and scores are invented for teaching.

Our first question concerns the students assigned to Massed practice: how do the three feedback-condition means compare? Chapter 3 developed an estimate, SE, interval and test for one comparison. We extend that reasoning to several groups, keeping the difference between an overall pattern and particular comparisons visible. Later, the full table lets us ask how the feedback comparison depends on practice schedule.

# 1 Comparison Questions and Designs

## 1.1 Outcome, factors, cells and units

A quiz score is the number of correctly answered items, from 0 to 20. Higher values represent more correct answers on this assessment. A four-point difference therefore means four more correct answers on average; it does not by itself describe every aspect of learning fractions. The outcome and its scale give a numerical comparison its substantive meaning.

A **factor** in analysis of variance is a categorical variable defining groups or conditions. Its categories are **levels**. Feedback has three levels and practice schedule has two. This use of *factor* differs from a latent factor used to represent shared indicator variation in a measurement model.

A **cell** contains cases sharing one level of each factor. Answer feedback with Spaced practice is one cell. Crossing three feedback levels with two practice levels gives six cells; four students

per cell makes the layout **balanced**. One-way and two-way refer to the number of factors, not to the number of levels. A one-way analysis can compare three, four or more groups.

**Table 1**

*Invented fractions-quiz scores, four different students per cell, on a 0–20 scale.*

| Feedback condition   | Massed practice | Spaced practice |
|----------------------|-----------------|-----------------|
| Rereading            | 4, 5, 7, 8      | 4, 5, 7, 8      |
| Answer feedback      | 6, 7, 9, 10     | 10, 11, 13, 14  |
| Explanatory feedback | 8, 9, 11, 12    | 16, 17, 19, 20  |

Each score in Table 1 belongs to a different student. The student is both the stipulated assignment unit and the measurement unit. The four scores in a cell are not repeated measurements on one student; nor are the Massed and Spaced scores paired. The repeated within-cell pattern deliberately simplifies the arithmetic. It is a teaching construction, not evidence of Normal errors, equal variances or an educational effect.

The complete course data file, `feedback_practice.csv`, represents the same table in one row per student, with columns `student`, `feedback`, `practice` and `score`. Students are numbered 1–24, ordered by feedback, then practice, then student within each cell. The printed scores are sufficient for every calculation in this chapter.

## 1.2 A one-way slice and the full factorial question

For the first analysis, restrict attention to the twelve Massed-practice students. There are three groups, defined by feedback alone. Their model means, written  $\mu_R$ ,  $\mu_A$  and  $\mu_E$ , refer to Rereading, Answer feedback and Explanatory feedback within this practice schedule. For example,  $\mu_A - \mu_R$  states a direction and an outcome scale. The observed difference estimates this particular contrast under the model introduced below.

This one-way analysis answers a prespecified question within a factorial study. Its conclusion concerns the Massed slice. It does not silently average over the Spaced students or establish that the same contrast holds in both schedules. Returning to all six cells will allow the second question: does the feedback comparison change with practice?

**Table 2***Two questions using stated parts of the same invented study.*

| Analysis | Records used                                    | Question addressed                                                                              |
|----------|-------------------------------------------------|-------------------------------------------------------------------------------------------------|
| One-way  | 12 Massed students; three feedback groups       | Do the feedback means differ within Massed practice?                                            |
| Two-way  | All 24 students; six feedback-by-practice cells | What are the average factor comparisons, and how do comparisons change across the other factor? |

A two-factor ANOVA and Chapter 3's two-way frequency table have different outcomes. ANOVA compares a numerical score across factor combinations. A frequency table counts cases in combinations of categories. The word *two-way* describes a layout; it does not identify a shared statistical test.

### 1.3 Design determines the relevant comparison

Random allocation is stipulated in the invented story. If implemented in a real study, that design could support a causal comparison among the participating students, subject to adherence, relevant interference and missing outcomes. It would not make the participants a random sample of all students. The data table alone establishes neither implementation of assignment nor representativeness.

Different designs can produce superficially similar tables. If whole classes receive conditions, students within the same class may share influences and the class may be the assignment unit. If each student is measured twice, the two outcomes belong to a pair. An uncertainty formula must reflect these connections rather than treating every row as an independent assignment.

**Blocking** makes condition comparisons within groups of similar units. Chapter 3's paired-school reference gives the complete illustration: each school contributes two classes, with one class assigned to each format. The school-specific differences preserve that allocation structure. The six cells in our present quiz example instead contain distinct individually assigned students. Sampling strata, which organize selection into a sample, and allocation blocks, which organize treatment comparisons, have different roles.

## 2 One-Way ANOVA

### 2.1 Inspect grouped observations before reducing them

The three Massed groups have means 6.0000, 8.0000 and 10.0000 points. Each has four scores and the same sample SD, 1.8257. Their ranges overlap: for example, some Rereading scores exceed some Answer-feedback scores. A difference between means is a summary of group distributions; it does not say that every student in one group outperformed every student in another.

**Table 3**

*Descriptions of the twelve Massed-practice scores.*

| Feedback             | n | Mean    | SD     | Range |
|----------------------|---|---------|--------|-------|
| Rereading            | 4 | 6.0000  | 1.8257 | 4–8   |
| Answer feedback      | 4 | 8.0000  | 1.8257 | 6–10  |
| Explanatory feedback | 4 | 10.0000 | 1.8257 | 8–12  |

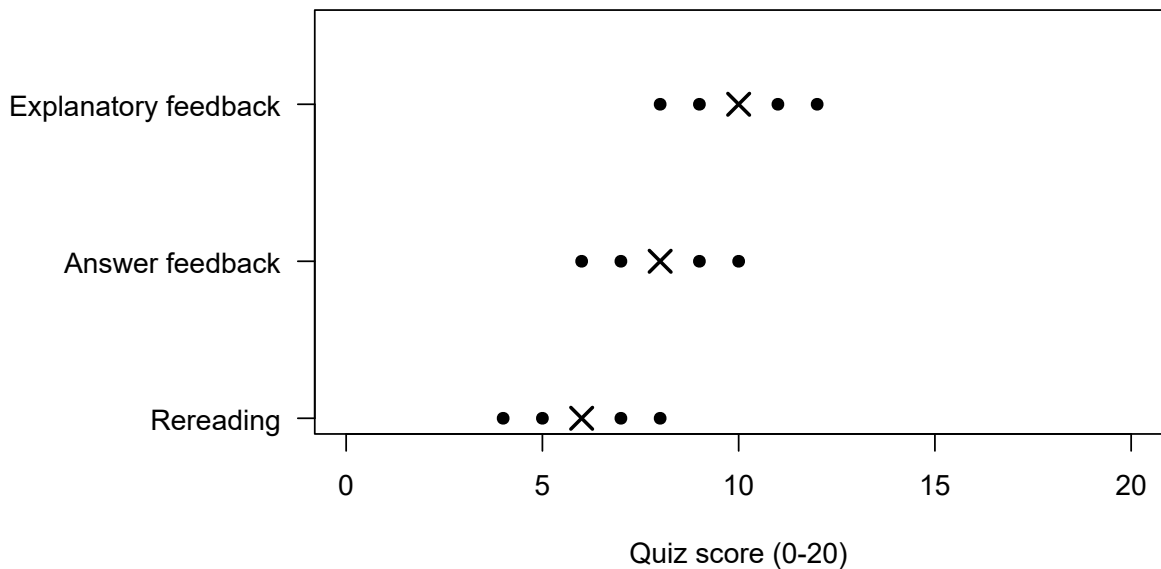


Figure 1: Massed-practice quiz scores by feedback condition. Every point is a different student; crosses mark group means.

With only four observations per group, individual dots communicate the record more directly than a boxplot. A bar chart of means would hide the within-group variation entirely. That

variation matters to inference: the same distance between group means can be more or less convincing depending on how widely individual scores vary within groups.

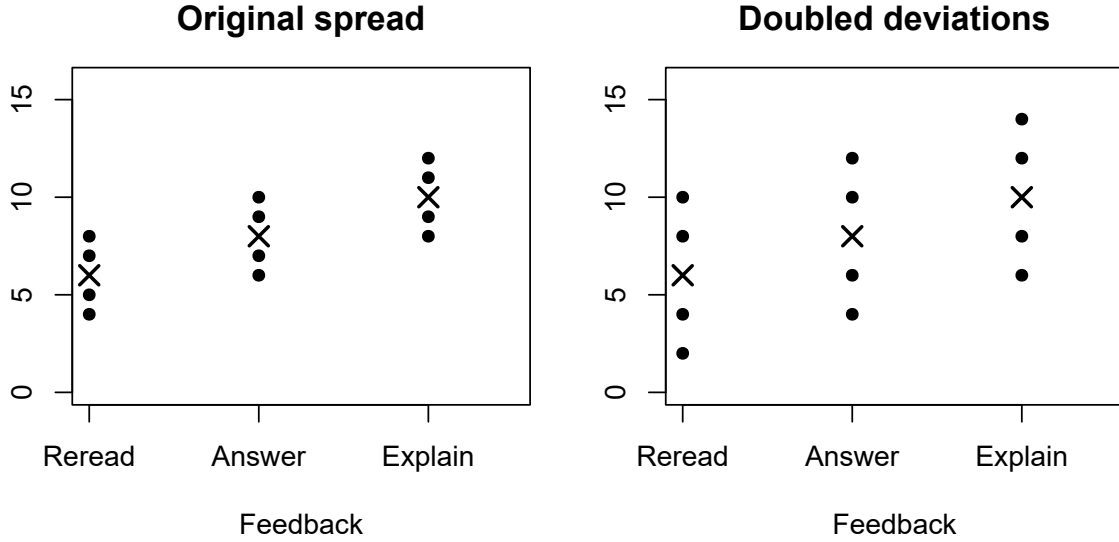


Figure 2: The same group means can provide different evidence when within-group spread changes. Counts and mean positions are held fixed in these constructed panels.

In Figure 2 the mean positions and counts stay fixed. Only each score’s distance from its group mean changes. Doubling those distances multiplies their squares by four. The second panel therefore has the same between-group separation but four times the within-group squared variation. ANOVA compares these two kinds of variation after putting them on comparable scales (Çetinkaya-Rundel & Hardin, 2024; Kutner et al., 2005).

## 2.2 Mean model and deviation decomposition

Write  $Y_{ij}$  for the random score of student  $i$  in feedback group  $j$ . Here  $j$  indexes a group, rather than a data-file column. The one-way model is

$$Y_{ij} = \mu_j + \varepsilon_{ij}.$$

The unknown  $\mu_j$  is the group mean in the model, and  $\varepsilon_{ij}$  is that student’s unknown departure from it. After fitting the group means, the **fitted value** is  $\hat{y}_{ij} = \bar{y}_j$ , and the **residual** is  $e_{ij} = y_{ij} - \bar{y}_j$ . A residual is calculated from observed scores and fitted means; a model error is a departure from an unknown model mean. This distinction will recur in regression.

The observed grand mean is  $\bar{y} = (6 + 8 + 10)/3 = 8.0000$  because the three groups are equally sized. For the Rereading student who scored 4, the distance from the grand mean separates into two parts:

$$4 - 8 = (6 - 8) + (4 - 6) = -2 + (-2).$$

The group mean is two points below the grand mean, and this student is another two points below the group mean. The general identity is

$$y_{ij} - \bar{y} = (\bar{y}_j - \bar{y}) + (y_{ij} - \bar{y}_j).$$

Squaring every deviation and adding over students gives a corresponding partition of total squared variation:

$$SS_T = SS_B + SS_W,$$

$$\begin{aligned} SS_T &= \sum_{j=1}^J \sum_{i=1}^{n_j} (y_{ij} - \bar{y})^2, \\ SS_B &= \sum_{j=1}^J n_j (\bar{y}_j - \bar{y})^2, \\ SS_W &= \sum_{j=1}^J \sum_{i=1}^{n_j} (y_{ij} - \bar{y}_j)^2. \end{aligned}$$

$SS_B$  is the **between-group sum of squares**: each group-mean departure is counted once for each student in that group.  $SS_W$  is the **within-group sum of squares**: the remaining squared residuals around those group means. The total  $SS_T$  describes departures around one grand mean. All three quantities are in squared quiz points.

Here the between-group calculation is

$$SS_B = 4(6 - 8)^2 + 4(8 - 8)^2 + 4(10 - 8)^2 = 32.0000.$$

Each group's residuals are  $-2, -1, 1, 2$ , whose squared sum is  $4 + 1 + 1 + 4 = 10$ . Thus

$$SS_W = 3(10) = 30.0000, \quad SS_T = 32 + 30 = 62.0000.$$

Squaring an individual deviation split introduces a cross-product. Those cross-products cancel only after summing within each group, because its residuals sum to zero. Appendix A shows that

cancellation explicitly. The decomposition is arithmetic; it needs no Normality or independence assumption. Those conditions enter when the observed decomposition is used for inference.

## 2.3 Degrees of freedom, mean squares and the F test

A sum of squares increases with both variation and the number of contributing observations. **Mean squares** divide by the appropriate degrees of freedom before the two sources are compared.

There are  $J = 3$  group-mean departures from the grand mean, but their weighted sum is zero. Only  $J - 1 = 2$  departures can vary freely. Within each group, four residuals sum to zero, leaving three free residuals. Across three groups this gives  $N - J = 12 - 3 = 9$  within-group df. The total has  $N - 1 = 11$  df, since all deviations from the grand mean sum to zero. The component df add:  $2 + 9 = 11$ .

$$MS_B = \frac{SS_B}{J-1} = \frac{32}{2} = 16.0000, \quad MS_W = \frac{SS_W}{N-J} = \frac{30}{9} = 3.3333.$$

The **F statistic** compares these mean squares:

$$F = \frac{MS_B}{MS_W} = \frac{16}{30/9} = 4.8000.$$

**Table 4**

*One-way ANOVA for the 12 Massed-practice students.*

| Source        | SS      | df | MS      | F      | p     |
|---------------|---------|----|---------|--------|-------|
| Feedback      | 32.0000 | 2  | 16.0000 | 4.8000 | .0381 |
| Within groups | 30.0000 | 9  | 3.3333  |        |       |
| Total         | 62.0000 | 11 |         |        |       |

The omnibus null hypothesis is

$$H_0 : \mu_R = \mu_A = \mu_E.$$

The alternative says that at least two of these means differ. Under independent observations with Normal errors and a common error variance  $\sigma^2$ , the null statistic has an  $F(2, 9)$  reference distribution. Its two df describe the numerator and denominator. Normality concerns the departures around the group means, not the pooled mixture of all scores.

Under the null, both mean squares estimate the same error variance. The numerator also responds to separation of the model means under alternatives. A relatively large ratio is therefore evidence against equality. The null does not force an observed ratio to equal 1, and the expectation of the ratio is not exactly 1. Sampling variation produces a distribution of possible positive ratios.

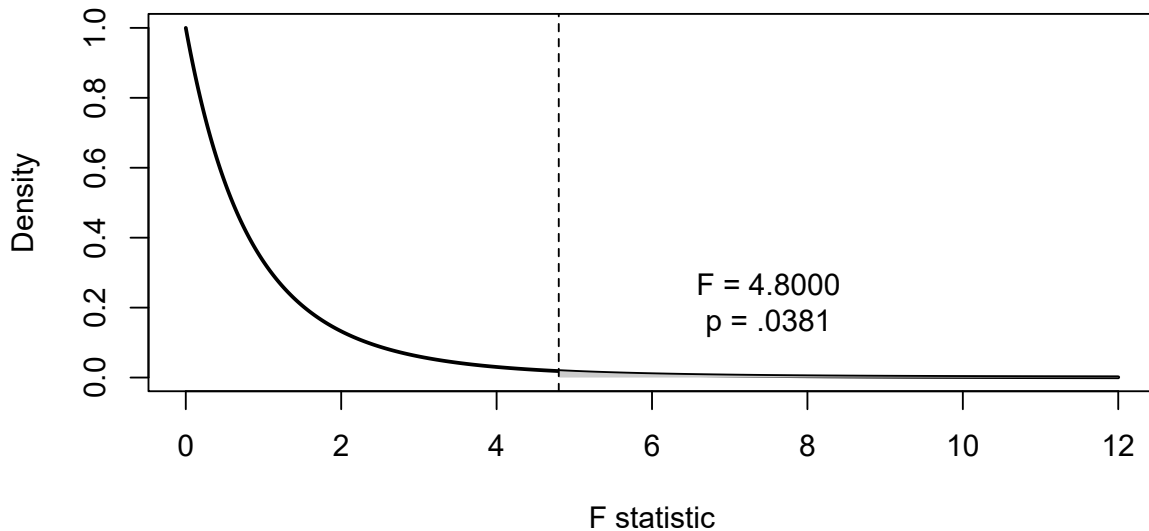


Figure 3: The one-way statistic 4.8000 and its right-tail probability under an  $F(2,9)$  null reference.

The right-tail probability in Figure 3 is .0381. Under the null and model conditions, a ratio at least as large as 4.8000 occurs with that probability. At a prespecified .05 significance level the equality null is rejected. This is evidence about the three-mean pattern, not a probability that the null is true and not an identification of which particular comparisons differ.

Statistical software fits this group-mean model from the twelve Massed records and reports the same table; the displayed arithmetic explains what that table means. The Computer Lab supplies the software routes, and Chapter 6 develops them further. A different software interface does not change the need to identify the records, factor levels and error model.

## 2.4 Read an ANOVA table

Every ANOVA table expresses one sentence about the observations:

$$\text{Data} = \text{fit} + \text{residual}.$$

The *fit* is what the estimated means account for, and the *residual* is what is left around them. Squaring and adding turns that sentence into the one-way partition already calculated, with a matching partition of degrees of freedom:

$$SS_T = SS_B + SS_W, \quad df_T = df_B + df_W,$$

$$62.0000 = 32.0000 + 30.0000, \quad 11 = 2 + 9.$$

A balanced two-factor layout keeps the same sentence and divides the fit part further, into two factor components and their interaction:

$$SS_T = SS_A + SS_B + SS_{AB} + SS_E, \quad df_T = df_A + df_B + df_{AB} + df_E.$$

For the complete study in Table 12, this reads  $476.0000 = 256.0000 + 96.0000 + 64.0000 + 60.0000$  and  $23 = 2 + 1 + 2 + 18$ . The fit part is therefore  $256 + 96 + 64 = 416.0000$  on  $2 + 1 + 2 = 5$  df. Both additivity checks are useful arithmetic checks on any printed table.

The compact six-column table used here keeps one row for each source of interest. Statistical software often prints a longer version of the same decomposition, adding rows that serve the fitting algorithm rather than the comparison question. Table 5 maps one layout onto the other.

**Table 5**

*How the compact table rows correspond to an extended software layout.*

| Extended output row | One-way value here           | Read or ignore                                                                                              |
|---------------------|------------------------------|-------------------------------------------------------------------------------------------------------------|
| Corrected Model     | 32.0000 on 2 df              | The whole fit part. With one factor it repeats the factor row; in the two-way layout it is 416.0000 on 5 df |
| Intercept           | A grand-mean term            | Ignore. It tests whether the overall mean is zero, which is not a comparison question for quiz scores       |
| Factor rows         | Feedback 32.0000 on 2 df     | Read. In the two-way layout these are 256.0000, 96.0000 and 64.0000 on 2, 1 and 2 df                        |
| Error               | 30.0000 on 9 df              | Read. It supplies $MS_W$ , or $MS_E$ , and the denominator df of every F ratio                              |
| Total               | Includes the grand-mean term | Ignore. This uncorrected total does not equal the sum of the rows above                                     |
| Corrected Total     | 62.0000 on 11 df             | Read. It equals the fit plus error rows, 476.0000 on 23 df in the two-way layout                            |

Read the factor rows, the error row and the corrected total; the intercept and uncorrected total answer no question asked here. The same reading order applies to the factorial table later in this chapter, where the interaction row is examined before the two main effects.

## 2.5 Two groups, direction and magnitude

For exactly two independent groups under the common-variance model, the classical pooled two-sample t test and one-way F test give

$$F(1, N - 2) = t_{N-2}^2.$$

The tests have the same p-value when the t alternative is two-sided. Squaring removes direction: positive and negative t values of the same magnitude give the same F. Keep the signed mean difference when explaining which group has the higher mean.

For a matched numerical example, use only the Massed Rereading and Answer groups. Their mean difference, Rereading minus Answer, is  $6 - 8 = -2.0000$ . Each sample variance is  $10/3$ , so the pooled variance is also  $10/3$ . These eight students give six residual df:

$$t(6) = \frac{-2}{\sqrt{(10/3)(1/4 + 1/4)}} = -1.5492, \quad F(1, 6) = 2.4000,$$

with the same two-sided  $p = .1723$ . This reduced fit differs from the three-group analysis, which had nine residual df. A selected contrast using that three-group error estimate is another calculation. The identity also does not equate a Welch t procedure, the default in some software, with a classical pooled ANOVA. Appendix A derives the matched identity and Chapter 3 explains the different calibrations.

Magnitude begins on the quiz scale. The observed Answer-minus-Rereading difference is 2.0000 points, and the Explanatory-minus-Rereading difference is 4.0000 points within Massed practice. Whether a difference of this size matters educationally needs a substantive criterion, not only a p-value.

One descriptive summary of the overall separation is **eta squared**:

$$\eta^2 = \frac{SS_B}{SS_T} = \frac{32}{62} = .5161.$$

Group means account for .5161 of the observed squared variation in these twelve records. This is a fraction of a sample decomposition, not the fraction of learning caused by feedback. It does not replace the actual group means or their pairwise differences. A standardized mean difference similarly needs its denominator stated: dividing a difference by an individual-score

SD produces a different quantity from dividing it by the SE of that difference. A bias-adjusted omnibus summary, omega squared, is described briefly in Appendix B.

### 3 Contrasts and Multiple Comparisons

#### 3.1 State the comparison through its weights

The omnibus question asks whether all three model means agree. A **contrast** states a particular comparison as a weighted combination of means whose weights sum to zero:

$$C = \sum_{j=1}^J c_j \mu_j, \quad \sum_{j=1}^J c_j = 0, \quad \widehat{C} = \sum_{j=1}^J c_j \bar{y}_j.$$

The symbol  $C$  here names a contrast, and  $\widehat{C}$  its estimate. The weights specify both the comparison and its direction. In the order Rereading, Answer feedback, Explanatory feedback:

**Table 6**

*Three contrast meanings within Massed practice.*

| Comparison                                                   | Weights          | Estimate              |
|--------------------------------------------------------------|------------------|-----------------------|
| Answer minus Rereading                                       | $(-1, 1, 0)$     | $-6 + 8 = 2.0000$     |
| Explanatory minus Rereading                                  | $(-1, 0, 1)$     | $-6 + 10 = 4.0000$    |
| Equal average of the two feedback approaches minus Rereading | $(-1, 1/2, 1/2)$ | $-6 + 4 + 5 = 3.0000$ |

The third comparison gives the two feedback approaches equal weight. It is not a substitute for both separate comparisons when both are scientifically required. In unequal groups, equal method weights and weights proportional to student counts answer different questions. The present groups have equal counts, so the two averaging rules coincide; Appendix B develops the distinction.

A grand mean with weights  $(1/3, 1/3, 1/3)$  is a useful linear combination but not a contrast, because those weights sum to 1. Multiplying every contrast weight by two doubles the estimate and SE. It preserves a zero-null t statistic, but changes the quantity being reported. Naming the comparison in ordinary words prevents a correct calculation from answering the wrong question.

### 3.2 Calculate the uncertainty for that comparison

Under the independent common-variance model, each estimated group mean has variance  $\sigma^2/n_j$ . The variance of a weighted sum uses the **squared weights**. Opposite signs define a difference; they do not make independent sources of uncertainty cancel. Replacing  $\sigma^2$  by the full three-group estimate  $MS_W$  gives

$$\widehat{SE}(\widehat{C}) = \sqrt{MS_W \sum_{j=1}^J \frac{c_j^2}{n_j}}.$$

For each Rereading-referenced difference,

$$\widehat{SE} = \sqrt{\frac{10}{3} \left( \frac{1}{4} + \frac{1}{4} \right)} = 1.2910 \text{ points.}$$

The three-group fit supplies nine error df even though each contrast uses two of its means. The error variance was estimated from all three groups under the common-variance assumption. This differs from refitting only two groups, as in the t/F identity above.

For the equal-average contrast, the squared weights give

$$\widehat{SE} = \sqrt{\frac{10}{3} \left( \frac{1}{4} + \frac{(1/2)^2}{4} + \frac{(1/2)^2}{4} \right)} = 1.1180.$$

This smaller SE belongs to a different estimand, an average of two means compared with the reference. It does not establish that either separate comparison is more precise than before. The NIST contrast explanation develops the same weighted-mean variance calculation.

### 3.3 Declare the family before choosing the critical value

Suppose the planned family contains exactly two two-sided comparisons: Answer minus Rereading and Explanatory minus Rereading. We want an interval procedure with at least 95% **simultaneous coverage** for both target contrasts under the model.

Bonferroni allocates the .05 error allowance across these two intervals, giving .025 to each interval and .0125 to each tail. Its critical value is therefore

$$t^* = t_{1-.05/(2 \times 2);9} = t_{.9875;9} = 2.6850.$$

Using unrounded values, each margin is  $2.685010847(1.290994449) = 3.4663$ . The intervals are  $\widehat{C} \pm t^* \widehat{SE}$ :

**Table 7***Bonferroni 95% simultaneous intervals for the two stated comparisons in Massed practice.*

| Comparison                  | Estimate | SE     | Family lower | Family upper |
|-----------------------------|----------|--------|--------------|--------------|
| Answer minus Rereading      | 2.0000   | 1.2910 | -1.4663      | 5.4663       |
| Explanatory minus Rereading | 4.0000   | 1.2910 | 0.5337       | 7.4663       |

The two estimates share the Rereading mean and are therefore related. Bonferroni's bound does not require their independence: the probability of at least one coverage failure is no larger than the sum of the two individual failure probabilities. The guarantee concerns repeated use of the whole stated procedure, not a 95% probability assigned to the two realized intervals. The NIST Bonferroni reference explains this joint-coverage bound.

**Tukey's method** answers a different family question: simultaneous coverage for all three pairwise mean differences. Its intervals and adjusted p-values for these data are:

**Table 8***Tukey 95% simultaneous intervals for all three pairs.*

| Pairwise comparison         | Estimate | Lower   | Upper  | Adjusted p |
|-----------------------------|----------|---------|--------|------------|
| Answer minus Rereading      | 2.0000   | -1.6045 | 5.6045 | .3150      |
| Explanatory minus Rereading | 4.0000   | 0.3955  | 7.6045 | .0310      |
| Explanatory minus Answer    | 2.0000   | -1.6045 | 5.6045 | .3150      |

The intervals differ from Bonferroni's because both the protected family and its calibration differ. Statistical software returns these all-pairs comparisons from the fitted one-way model; the Computer Lab supplies the software routes. Dunnett's method addresses comparisons with a common reference; Scheffé's method can protect all contrasts in a specified mean-comparison space. The choice comes from the comparisons for which coverage is owed, not from which method happens to exclude zero (Kutner et al., 2005; Tukey, 1953).

A contrast is a mathematical form. It can be defined before relevant results are seen or selected after examining them. Selection matters because looking across possibilities can favor unusually large observed differences. A reported selected comparison needs uncertainty appropriate to that selection. Nor is rejection of an omnibus test a universal prerequisite for a prespecified comparison or a simultaneous interval. State the question, the family, the selection process and the calibration.

### 3.4 Read the omnibus and particular results together

The three-group F test concerns equality of all three model means. The two Bonferroni intervals concern the two narrower reference comparisons. The Explanatory-minus-Rereading interval excludes zero; the Answer-minus-Rereading interval includes zero. The latter remains compatible with both negative and positive differences and does not establish equivalence. No practical equivalence margin was declared.

A connected report would say that, in the Massed slice, observed means were 6.0000, 8.0000 and 10.0000 points. The omnibus common-variance Normal-model test gave  $F(2, 9) = 4.8000$ ,  $p = .0381$ . For the two declared reference comparisons, the estimates were 2.0000 and 4.0000 points, with simultaneous intervals  $[-1.4663, 5.4663]$  and  $[0.5337, 7.4663]$ . These statements preserve magnitude, uncertainty, family and the restriction to Massed practice. They describe calculations for invented scores, not an empirical educational finding.

## 4 Two-Way ANOVA

### 4.1 What interaction means, and why study two factors at once

Two factors **interact** when the effect of one factor differs across the levels of the other. Nothing more technical is required to state the idea. A combination of levels then produces a result that the two separate comparisons would not have led us to expect.

Here is one invented illustration, outside the quiz study. Suppose a short vocabulary preview is given before a reading passage, and that students are also grouped by whether they began with weak or strong vocabulary. The preview raises comprehension scores substantially for the students who began weak, and barely changes them for the students who began strong. The effect of the preview *depends on* prior vocabulary: that dependence is the interaction. Describing only an average preview effect would tell a reader that the preview helps somewhat, and would conceal that it helps one group a great deal and the other hardly at all.

Studying two factors together in one crossed design has three advantages over running two separate one-factor studies.

1. **It is more efficient.** Every student contributes to both factor comparisons at once, because the levels of one factor are spread across the levels of the other. A crossed design therefore needs fewer cases per condition than a series of separate one-factor studies aimed at the same comparisons.
2. **It reduces residual variation.** A second factor that influences the outcome moves variation out of the error term and into the fit. If all 24 students had been analyzed with feedback as the only factor, the practice and interaction components, 96.0000 and 64.0000, would have stayed inside the residual term instead of leaving the 60.0000 within-cell error used below.

3. **It makes interaction examinable.** Only a crossed design supports the question of whether a comparison changes across the other factor's levels. A series of one-factor studies supplies no estimate of that change, however carefully each one is run.

Adding practice schedule to the quiz study therefore gives each comparison a context. We can compare feedback conditions within Massed practice, within Spaced practice, or after averaging over the two schedules. These are distinct quantities. A **main effect** compares one factor's levels after averaging across the other factor with stated weights. A **simple effect** makes the comparison at a specified level of the other factor. **Interaction** means that a comparison changes across those levels.

## 4.2 Main effects and interaction through graphs

In a two-factor design either main effect, the interaction, or any combination of the three can be present. Six schematic two-by-two patterns make the whole range visible before we return to the full three-by-two data. Each line in Figure 4 represents a feedback level, A or B; the horizontal positions represent Massed and Spaced practice. These are mean-only drawings, not additional observed studies, and they carry no significance claim.

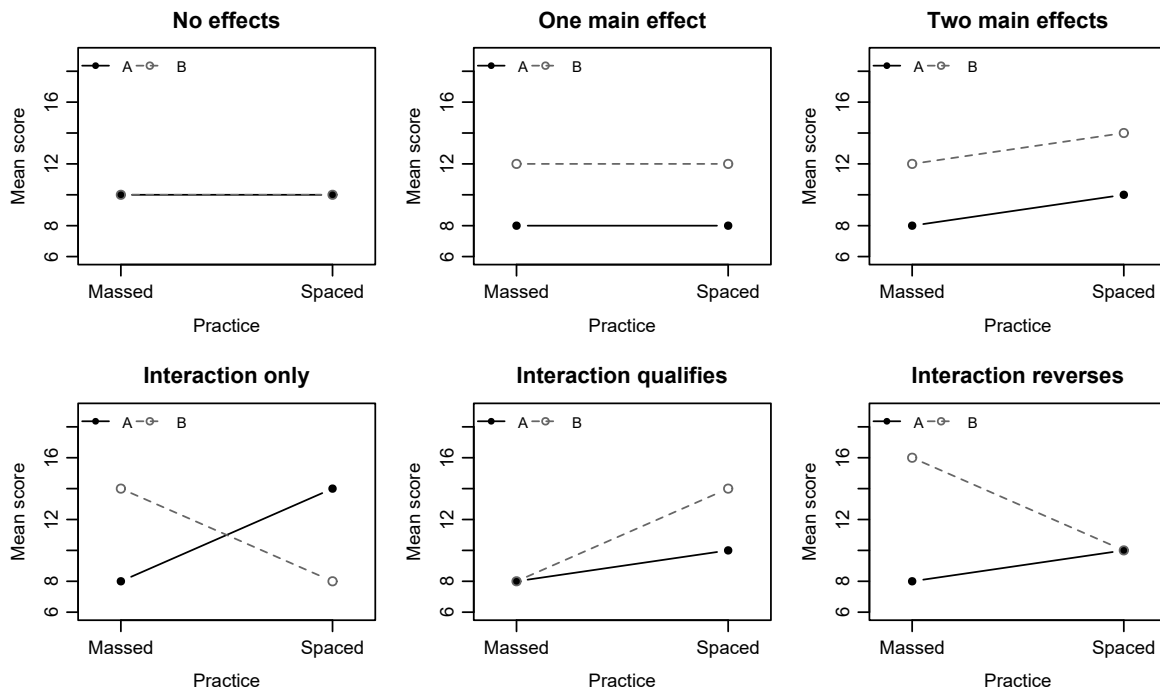


Figure 4: Six schematic patterns of cell means, covering the whole range of main-effect and interaction combinations. Lines identify feedback levels A and B; the drawings illustrate averages and changing differences, without significance claims.

Read each pattern by taking differences, not only by recognizing a shape. Define a schedule difference as Spaced minus Massed and a difference of differences as B's schedule difference minus A's.

**Table 9**

*Arithmetic behind the six schematic patterns.*

| Pattern               | A: Massed,<br>Spaced | B: Massed,<br>Spaced | Schedule<br>differences A, B | Difference of<br>differences |
|-----------------------|----------------------|----------------------|------------------------------|------------------------------|
| No effects            | 10, 10               | 10, 10               | 0, 0                         | 0                            |
| One main effect       | 8, 8                 | 12, 12               | 0, 0                         | 0                            |
| Two main effects      | 8, 10                | 12, 14               | 2, 2                         | 0                            |
| Interaction only      | 8, 14                | 14, 8                | 6, -6                        | -12                          |
| Interaction qualifies | 8, 10                | 8, 14                | 2, 6                         | 4                            |
| Interaction reverses  | 8, 10                | 16, 10               | 2, -6                        | -8                           |

In the first panel nothing varies: both feedback levels and both schedules have mean 10, and the two lines coincide. In the second, B exceeds A by 4 points under either schedule while the schedule comparison is 0 under either feedback level; there is one main effect and no interaction. In the third, B still exceeds A by 4 points everywhere, and Spaced now exceeds Massed by 2 points everywhere: two main effects, again with a zero difference of differences. Parallel lines express those constant comparisons.

The last three panels have a nonzero difference of differences. In the crossing panel, both feedback marginal means are 11 and both practice marginal means are 11 when cells receive equal weight. Both averaged comparisons are zero despite large conditional differences, which is why a small main effect need not mean that a factor is irrelevant at every level of another factor.

In the qualifying panel the schedule comparison is 2 for A and 6 for B. Both are positive, so the averaged 4-point schedule comparison points the right way for both feedback levels but describes neither exactly. The lines need not cross for interaction to be present. In the reversing panel the schedule comparison is 2 for A and -6 for B, so the averaged comparison of -2 points the wrong way for A. Reporting only the two main effects would then mislead a reader about every conditional comparison. Examine the interaction first, and read a main effect as an average whose parts are still visible.

These drawings establish only the numerical patterns: significance also needs information about variation, sample size and the reference distribution.

### 4.3 Cells, marginal means and the full data

Return to all 24 quiz scores. Each of the six cells contains four different students; no person contributes a score under the other practice schedule. The first one-way analysis used one three-cell slice, whereas the full analysis uses all six groups.

Write  $\bar{y}_{jk}$  for a cell mean, with  $j = 1, 2, 3$  indexing feedback and  $k = 1, 2$  indexing practice. A dot replaces an index that has been averaged over:  $\bar{y}_{j\cdot}$  averages a feedback level over practice, and  $\bar{y}_{\cdot k}$  averages a practice level over feedback.

**Table 10**

*Cell and equally weighted marginal means for the complete invented study.*

| Feedback             | Massed  | Spaced  | Feedback mean      |
|----------------------|---------|---------|--------------------|
| Rereading            | 6.0000  | 6.0000  | 6.0000             |
| Answer feedback      | 8.0000  | 12.0000 | 10.0000            |
| Explanatory feedback | 10.0000 | 18.0000 | 14.0000            |
| Practice mean        | 8.0000  | 12.0000 | Grand mean 10.0000 |

Answer feedback averages  $(8 + 12)/2 = 10.0000$  over schedules. Since cell sizes are equal, this equals the mean of its eight student scores. The Massed marginal mean is  $(6+8+10)/3 = 8.0000$ , and the Spaced marginal mean is  $(6 + 12 + 18)/3 = 12.0000$ . The grand mean is 10.0000.

The practice comparison averages  $12 - 8 = 4.0000$  points across the three feedback conditions with equal weights. Its component schedule differences are 0.0000, 4.0000 and 8.0000. A four-point main effect is therefore a meaningful stated average, but it describes none of the conditional comparisons except Answer feedback's.

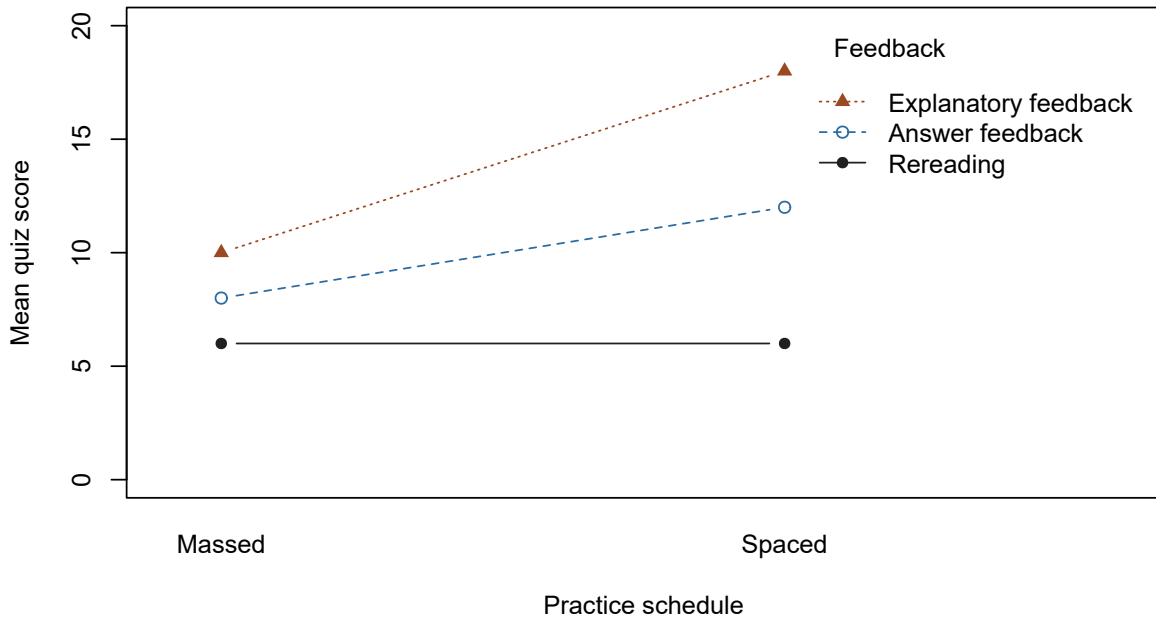


Figure 5: Quiz cell means across practice schedules in the full  $3 \times 2$  example. Each line joins means of different students; the schedule differences are 0, 4 and 8 points.

The connected lines in Figure 5 organize cell means. They do not show individual students moving from one condition to another. The nonparallel pattern describes interaction in the observed means; the full model supplies the uncertainty calculation for an inferential claim.

#### 4.4 Additivity and the full model

An additive prediction combines the grand mean with the two marginal departures. For Answer-Spaced, it is

$$10 + (10 - 10) + (12 - 10) = 12,$$

which agrees with that observed cell mean. For Explanatory-Spaced, the additive prediction is

$$10 + (14 - 10) + (12 - 10) = 16,$$

while the cell mean is 18. The extra 2 points are an **interaction departure** from that additive prediction. Agreement for one cell does not imply that the entire layout is additive.

The full mean model includes such departures:

$$Y_{ijk} = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk} + \varepsilon_{ijk}.$$

Here  $i = 1, \dots, 4$  identifies a student within a cell,  $j = 1, 2, 3$  identifies feedback and  $k = 1, 2$  identifies practice. The grand model mean is  $\mu$ ;  $\alpha_j$  and  $\beta_k$  are the factor effects;  $(\alpha\beta)_{jk}$  is the interaction departure; and  $\varepsilon_{ijk}$  is the individual error around the cell mean. These indexed effects are distinct from the unindexed error probabilities  $\alpha$  and  $\beta$  used in hypothesis testing.

In the balanced decomposition, main-effect departures sum to zero. Interaction departures sum to zero along each row and each column, so they contain only the remaining pattern after the two main effects. The observed feedback departures are  $-4, 0, 4$ , and practice departures are  $-2, 2$ . The interaction departures are:

**Table 11**

*Observed interaction departures from the additive prediction.*

| Feedback             | Massed | Spaced |
|----------------------|--------|--------|
| Rereading            | 2      | -2     |
| Answer feedback      | 0      | 0      |
| Explanatory feedback | -2     | 2      |

With two columns, choosing a departure in the first column fixes the opposite departure in the second. Of the three first-column departures, only two are free because they sum to zero. Thus the interaction has  $(3 - 1)(2 - 1) = 2$  df. It describes a two-dimensional pattern, rather than one particular four-cell difference.

For a common cell size  $n = 4$ , the total squared variation splits as

$$SS_T = SS_A + SS_B + SS_{AB} + SS_E.$$

In this **two-way** table, A denotes feedback and B denotes practice. Thus  $SS_B$  now means the practice component, not the entire between-group component of the one-way table.  $SS_E$  is the within-cell error sum of squares. Stating the local meanings keeps the two familiar uses of B distinct.

$$\begin{aligned}
SS_A &= 2n \sum_j (\bar{y}_{j\cdot} - \bar{y})^2 = 256.0000, \\
SS_B &= 3n \sum_k (\bar{y}_{\cdot k} - \bar{y})^2 = 96.0000, \\
SS_{AB} &= n \sum_j \sum_k (\bar{y}_{jk} - \bar{y}_{j\cdot} - \bar{y}_{\cdot k} + \bar{y})^2 = 64.0000, \\
SS_E &= \sum_j \sum_k \sum_i (y_{ijk} - \bar{y}_{jk})^2 = 60.0000.
\end{aligned}$$

For example, the feedback component counts each marginal departure eight times:  $8\{(-4)^2 + 0^2 + 4^2\} = 256$ . The practice component counts each departure twelve times:  $12\{(-2)^2 + 2^2\} = 96$ . Four students share each interaction departure, giving  $4(4 + 4 + 0 + 0 + 4 + 4) = 64$ . Each cell contributes 10 residual squared points, giving  $6(10) = 60$ . Their sum is  $SS_T = 476$ .

Each of six cells has an estimated mean, so error df are  $24 - 6 = 18$ . Feedback has two df, practice has one, and interaction has two. Together they give  $2 + 1 + 2 + 18 = 23 = N - 1$ .

**Table 12**

*Full two-way ANOVA using all 24 students.*

| Source       | SS       | df | MS       | F       | p      |
|--------------|----------|----|----------|---------|--------|
| Feedback (A) | 256.0000 | 2  | 128.0000 | 38.4000 | <.0001 |
| Practice (B) | 96.0000  | 1  | 96.0000  | 28.8000 | <.0001 |
| A $\times$ B | 64.0000  | 2  | 32.0000  | 9.6000  | .0015  |
| Error        | 60.0000  | 18 | 3.3333   |         |        |
| Total        | 476.0000 | 23 |          |         |        |

Read this table with the reading order set out in Read an ANOVA table: the three source rows, the error row and the corrected total. An extended software layout prints the same components with a Corrected Model row of 416.0000 on 5 df, plus an intercept row and an uncorrected total that answer no question asked here.

Every displayed F divides its source mean square by the same  $MS_E = 60/18 = 10/3$ . The corresponding null hypotheses address different features of the model:

**Table 13***Three null hypotheses with their averaging rules.*

| Source      | Null hypothesis                         | Comparison represented                                                                                                          |
|-------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Feedback    | $\mu_{R.} = \mu_{A.} = \mu_{E.}$        | Equal feedback means, each averaging the two schedules equally                                                                  |
| Practice    | $\mu_{.M} = \mu_{.S}$                   | Equal schedule means, each averaging the three feedback levels equally                                                          |
| Interaction | $(\alpha\beta)_{jk} = 0$ for every cell | Each feedback comparison is constant across schedules, equivalently each schedule difference is the same across feedback levels |

Independent common-variance Normal errors supply the classical F references. Under these conditions, the interaction result  $F(2, 18) = 9.6000$ ,  $p = .0015$  gives evidence against an additive pattern. Both main effects also have small p-values, but their meanings remain averaged comparisons. Reading them as “each factor works equally well everywhere” would discard the interaction.

The clean component decomposition uses the balanced layout. With unequal cell counts, the averaging target and the model comparison need more care; different software sums of squares can address different hypotheses. The weighting reference explains the first distinction without changing these data.

## 4.5 Simple effects and a selected interaction contrast

The Answer-feedback schedule comparison is a **simple effect**:  $12 - 8 = 4.0000$  points. Its SE under the full model is

$$\widehat{SE} = \sqrt{\frac{10}{3} \left( \frac{1}{4} + \frac{1}{4} \right)} = 1.2910.$$

With 18 residual df, a pointwise 95% interval is

$$4 \pm t_{.975, 18} \sqrt{5/3} \approx [1.2877, 6.7123].$$

This interval is calibrated for the one stated simple comparison. It is not protected for every simple effect that a reader might inspect after seeing the graph. The two means come from different students, so this is an independent-cell calculation, not a paired calculation.

Now suppose a reader selects the Answer-versus-Rereading change in the schedule comparison after inspecting the pattern. In model means, the target is

$$C = (\mu_{A,S} - \mu_{A,M}) - (\mu_{R,S} - \mu_{R,M}).$$

Its estimate is one **difference of differences**:

$$\widehat{C} = (12 - 8) - (6 - 6) = 4.0000.$$

The four independent cell means have weights 1, -1, -1, 1 in the displayed expression. Each has variance estimated by  $MS_E/4$ ; squaring the weights and adding gives

$$\widehat{SE}(\widehat{C}) = \sqrt{\frac{10}{3} \left( \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} \right)} = 1.8257.$$

To account for selection among interaction contrasts, choose Scheffé coverage for **all contrasts in the two-dimensional interaction space**. The multiplier is  $\sqrt{2F_{.95;2,18}} = 2.6663$ . The complete simultaneous calculation is

$$\begin{aligned} \widehat{C} \pm \sqrt{2F_{.95;2,18}} \widehat{SE}(\widehat{C}) &= 4 \pm 2.666292237 \sqrt{10/3} \\ &\approx [-0.8680, 8.8680]. \end{aligned}$$

The interval includes zero even though the omnibus interaction p-value is .0015. There is no contradiction. The joint test assesses the entire two-dimensional departure from additivity; the selected contrast concerns one direction, with protection for choosing among all interaction contrasts. Evidence that some interaction direction is nonzero does not establish every selected direction.

The simple Answer schedule effect includes the practice main-effect component as well as an interaction component. It therefore lies outside this interaction-only Scheffé family. A claim covering all simple effects would need a corresponding family and calibration. Appendix B compares the selected contrast's pointwise and simultaneous intervals using the same estimate and SE.

## 4.6 Give a connected interpretation

The observed schedule differences are 0.0000, 4.0000 and 8.0000 quiz points for Rereading, Answer feedback and Explanatory feedback. The full-model interaction test gives  $F(2, 18) = 9.6000$ ,  $p = .0015$  under the stated conditions. The selected Answer-versus-Rereading change in schedule difference is 4.0000 points, with a 95% Scheffé interval  $[-0.8680, 8.8680]$  over all interaction contrasts. The overall result does not establish that particular selected comparison.

Feedback and practice marginal means summarize explicit equal weights. The cell means show what those averages can conceal. The one-way conclusion concerns only the Massed students, whereas the full factorial conclusion uses all 24 students and 18 error df. These are analyses of invented data, not empirical evidence that a teaching method succeeds in a population of students.

## 5 Conditions and Interpretation

### 5.1 Explain the conditions in familiar terms

Three conditions support the classical ANOVA references used here: **independent observations**, **similar variances within groups or cells**, and **approximately Normal residuals**. More precisely, the exact finite-sample F and t results assume independent Normal errors with a common variance around the relevant model means. Residuals provide observable clues about those errors, together with the design and substantive knowledge (Kutner et al., 2005).

Independence is principally a design question. It concerns how outcomes can share influences, including repeated observations, classroom clustering or interference between students. A residual plot cannot establish independent assignment or independent errors. Similar within-cell spread is a different question: does a common error variance reasonably describe each cell? The pooled spread of all outcomes need not be small or unimodal, because the groups can have different means.

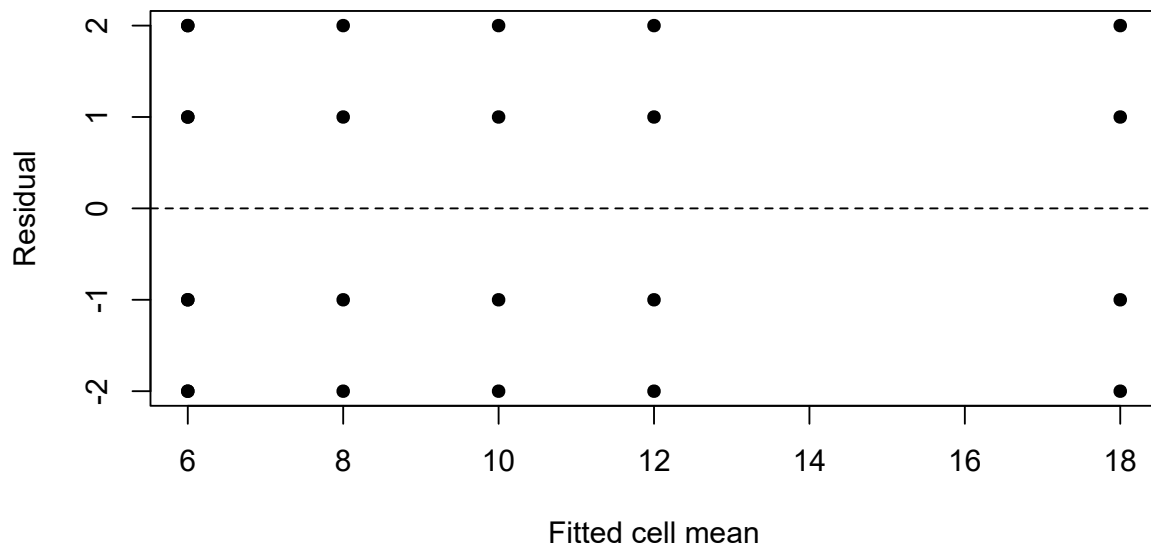


Figure 6: Residuals around the fitted cell means. The repeated vertical pattern was deliberately built into the invented scores; it is not independent evidence that the model conditions hold.

A residual-versus-fitted display can reveal changing spread or an unusual residual. Here each cell has residuals  $-2, -1, 1, 2$ , because we deliberately constructed the scores that way. Four students per cell provide very little information for judging a distribution's tails. Identical plotted spreads in this example do not validate a Normal model for real quiz scores.

A Normal quantile plot compares ordered residuals with quantiles of a Normal distribution. It can make pronounced tail departures more visible, but a small apparently straight display offers limited reassurance. Treat such graphs as information to interpret; a sequence of Normality or variance tests should not become an automatic permission rule for ANOVA.

**Table 14***Conditions connect design and observed patterns to interpretation.*

| Question                                       | Information to examine                                             | Implication for the calculation                                              |
|------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------|
| Are observations independent?                  | Assignment unit, repeated measurements, classes and schools        | Dependence changes the covariance and relevant error structure               |
| Is a common variance reasonable?               | Group/cell spreads, residual displays and subject-matter knowledge | Unequal-variance procedures use a different uncertainty calibration          |
| Is the error shape adequate for the reference? | Residual distribution, unusual observations and cell sizes         | Small samples can be sensitive to departures from the stated reference model |
| Does the comparison fit the question?          | Outcome, factor order, selected records and averaging weights      | A correct calculation can otherwise describe a different target              |

Welch's one-way method allows unequal group variances and changes the uncertainty calibration. It is not obtained by swapping one number in the classical sum-of-squares table. It does not solve clustering, and its one-way form does not supply the full factorial interaction analysis. Chapter 6 shows the alternative alongside the common-variance calculation, with its conditions kept visible.

## 5.2 Missing outcomes and the scope of the answer

A report should compare assigned and analyzed counts and explain known reasons for losses. Missing outcomes can alter who contributes to a comparison and can require assumptions beyond the displayed ANOVA model. Our invented dataset has all 24 scores, so it supplies no missingness experiment or evidence about a deletion rule. In a real study, silently analyzing only available records would need justification at the intended target.

Keep three claims distinct. The observed means and sums of squares describe these records. Intervals and tests express uncertainty under a specified model or design-based approximation. Causal and population interpretations require support from allocation, measurement, participation and the intended scope. A small p-value does not complete that support, and a large p-value does not prove that an effect is absent or unimportant.

## 6 Reading Bridge

Comparing several means extends the estimate-and-SE reasoning of Chapter 3. ANOVA separates sources of variation to ask an omnibus question; a contrast states a narrower question and its uncertainty. Main effects average across factor levels, while cell means and simple effects show the conditional pattern. A joint interaction test and one selected interaction contrast can therefore lead to different, compatible conclusions.

The same group-mean model can be written using indicator variables. A measured baseline can also enter an adjusted model, called analysis of covariance (ANCOVA). Chapter 6 gives their first complete applications, with stated comparison targets and conditions. Chapters 7–8 develop the regression mathematics, several controls and standardized predictions. Baseline adjustment accounts for a measured source of score variation and compares fitted outcomes at a stated common baseline value, or averages them over the same stated baseline distribution. The groups need adequate overlap in those baseline values, often called common support, for those comparisons to be informed by observations. Adjustment is not a general cure for confounding or missing outcomes.

Blocking organizes comparisons within allocation groups, as in Chapter 3’s paired-school reference. Repeated measurements and classroom clustering change the covariance structure and the relevant error term. The shape of a data table cannot settle these design questions. Chapters 5–6 now make the statistical foundations executable while keeping the question, records and interpretation beside each calculation.

### Key terms

Use this reference to revisit the concepts principally explained here. The Part One introduction collects earlier terms, including estimand, SE, confidence interval and familywise error rate.

**Table 15**

*Selected concepts for group comparisons and ANOVA.*

| Term and common alternatives | Meaning                                                                                                                                                                               |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Factor (grouping factor)     | In ANOVA, a categorical variable defining groups or conditions; its categories are levels. This meaning differs from a latent factor used to model shared indicator variation.        |
| Analysis of variance (ANOVA) | A method for studying mean differences through sources of variation. One-way and two-way ANOVA use one and two grouping factors, respectively; either factor can have several levels. |
| Fitted value                 | The value a fitted model assigns to an observed case. In a one-way group-mean model, it is that case’s group mean.                                                                    |

| Term and common alternatives      | Meaning                                                                                                                                                                                                                    |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Residual                          | An observed outcome minus its fitted value. A residual is calculated from the fitted model; an error is the corresponding unknown departure in the population model.                                                       |
| Mean square (MS)                  | A sum of squares divided by its degrees of freedom. In classical ANOVA, the within-group mean square estimates the common error variance under the model's conditions.                                                     |
| F statistic (F ratio)             | In classical ANOVA, a ratio of the mean square for a tested source to the appropriate error mean square. Its null F reference depends on the model and both degrees of freedom.                                            |
| Omnibus test                      | A test of one joint null hypothesis about a whole pattern, such as equality of all group means. Rejecting it does not identify which comparisons differ or how large they are.                                             |
| Standardized mean difference      | A difference in means divided by a specified SD. The chosen denominator determines the standardized quantity; the raw difference retains the outcome's original units.                                                     |
| Eta squared                       | The proportion of total observed variation attributed to a specified source in an ANOVA decomposition. In one-way ANOVA, it is between-group divided by total sum of squares.                                              |
| Contrast (linear contrast)        | A weighted combination of means whose weights sum to zero, expressing a specific comparison. The means, weights, and target population determine the quantity compared.                                                    |
| Simultaneous confidence intervals | A set of intervals calibrated so that, under their conditions, all of them cover their estimands together at the stated familywise rate; Bonferroni, Tukey, and Dunnett are different calibrations for different families. |
| Cell mean and marginal mean       | In a two-factor layout, a cell mean is the mean of the cases sharing one level of each factor; a marginal mean averages the cell means across the levels of the other factor, with stated weights.                         |
| Main effect                       | A comparison across one factor's levels averaged over the other factor's levels, using stated weights. It remains an average when the conditional pattern differs across levels.                                           |
| Interaction                       | A pattern in which a comparison across one factor's levels changes with the other factor's level. A difference of differences expresses one such comparison.                                                               |
| Simple effect                     | A comparison across one factor's levels at a specified level of the other factor. It describes a conditional comparison within the factorial design.                                                                       |

## Further reading

- *Introduction to Modern Statistics* develops the transition from grouped observations to inference for many means (Çetinkaya-Rundel & Hardin, 2024).
- Penn State STAT 200, Lesson 10 provides an introductory ANOVA treatment; its factorial-design lesson develops main effects and interaction.
- The NIST handbook explains contrasts and their SEs and Bonferroni simultaneous intervals.

## Exercises

The supplied scores and constants support calculator-based work. Retain full precision until reporting calculated results to four decimal places. State the relevant records, conditions and comparison direction in each interpretation.

### Exercise 1: Read grouped observations (Foundational)

Use the twelve Massed-practice scores from the invented study:

- Rereading: 4, 5, 7, 8.
  - Answer feedback: 6, 7, 9, 10.
  - Explanatory feedback: 8, 9, 11, 12.
1. Calculate each mean, range and sample SD. Compare the centers, spread and overlap. Use the sample-variance denominator  $n_j - 1$ .
  2. Draw or describe a dotplot displaying every score. Explain what it reveals that a bar chart showing only the means would conceal.
  3. Explain why the identical within-group spreads do not establish that a common-variance Normal model will hold in a real study.
  4. Identify the assignment and measurement units stipulated in this invented study. Explain what the scores alone cannot establish about causal interpretation or generalization.

### Exercise 2: Reproduce the one-way calculation (Foundational)

Use Exercise 1's scores. There are  $J = 3$  groups and  $N = 12$  students. The group means are 6, 8 and 10 and the grand mean is 8. For the classical test, assume independent Normal errors with a common variance.

1. For the Rereading score of 4, split its deviation from the grand mean into a group-mean departure and a residual. Calculate the squared total deviation, the two squared components and their cross-product. Explain why the sum-of-squares identity applies after summing, rather than to those two squares alone for every individual.
2. Calculate  $SS_B$ ,  $SS_W$  and  $SS_T$  from the scores. Explain why the within-group contribution is the same for all three groups.
3. Find all three df, both mean squares and the F statistic. Explain the constraint behind each df and what the ratio compares.
4. State the omnibus null. Interpret the supplied right-tail  $p = .0381$  at  $\alpha = .05$ , including the model conditions and what the conclusion leaves unanswered about particular comparisons.

### Exercise 3: Define and estimate contrasts (Foundational)

The Massed group means are 6, 8 and 10 in the order Rereading, Answer feedback, Explanatory feedback. Each group has four students and the common-variance estimate is  $MS_W = 10/3$ , with nine residual df.

1. Identify which weight vectors define contrasts:  $(-1, 1, 0)$ ,  $(-1, 0, 1)$ ,  $(-1, 1/2, 1/2)$  and  $(1/3, 1/3, 1/3)$ . State the ordinary-language question represented by each contrast.
2. Calculate each contrast estimate and its SE from  $\sqrt{MS_W \sum_j c_j^2/n_j}$ . Explain the squared weights and the use of the full three-group error estimate.
3. Explain why the equal-average contrast cannot replace both separate feedback-versus-Rereading comparisons when both are required.
4. For a weighting illustration only, suppose Answer and Explanatory means stay at 8 and 10 but their counts are 10 and 30. The Rereading mean stays at 6. Compare the equally weighted method-average difference with the student-weighted difference. State the question answered by each. These illustrative counts do not change the chapter's actual balanced dataset.

### Exercise 4: Interpret comparison families (Integrative)

The declared simultaneous family contains Answer minus Rereading and Explanatory minus Rereading in Massed practice. Their estimates are 2 and 4, each with  $SE = \sqrt{5/3}$  and nine error df. Use the Bonferroni critical value 2.685010847 under the independent common-variance Normal model.

1. Divide a .05 familywise error allowance between the two intervals and their tails. Explain the resulting quantile .9875.
2. Calculate the two simultaneous intervals and interpret them together. Explain why an interval containing zero does not establish equivalence.
3. Tukey's all-pairs intervals are  $[-1.6045, 5.6045]$  for Answer minus Rereading,  $[0.3955, 7.6045]$  for Explanatory minus Rereading, and  $[-1.6045, 5.6045]$  for Explanatory minus Answer. Identify its family and explain why its limits differ from those in part 2. Give a reason for choosing a family before inspecting which intervals exclude zero.
4. Both reference comparisons use the same Rereading mean. Explain why this creates dependence and why it does not invalidate the Bonferroni bound.
5. Distinguish an omnibus rejection, a prespecified contrast and a contrast selected after inspecting results. State what should accompany uncertainty reported for the selected contrast.

### Exercise 5: Read the factorial pattern (Foundational)

Use the complete invented study below. There are four different students per cell, individually allocated to the six combinations.

| Feedback             | Massed scores | Spaced scores  |
|----------------------|---------------|----------------|
| Rereading            | 4, 5, 7, 8    | 4, 5, 7, 8     |
| Answer feedback      | 6, 7, 9, 10   | 10, 11, 13, 14 |
| Explanatory feedback | 8, 9, 11, 12  | 16, 17, 19, 20 |

1. Identify the outcome, the two factors and their levels, the assignment unit, the number of cells and the meaning of balance.
2. Calculate the Answer-Spaced cell mean, the Answer marginal mean, both practice marginal means and the grand mean. Explain why equal-cell and student-weighted averages coincide here.
3. Calculate the additive prediction for Explanatory-Spaced from the grand mean and the two marginal departures. Compare it with the observed cell mean and identify the interaction departure.
4. Calculate Spaced minus Massed for each feedback condition. Then calculate Answer's schedule difference minus Rereading's. Explain how these values appear in the cell-mean plot.
5. Explain why the four-cell difference in part 4 is one interaction contrast, while the full three-by-two interaction has two df. Explain why the connected lines do not represent paired students.

### Exercise 6: Distinguish joint and selected results (Integrative)

For the full feedback-by-practice model, the feedback, practice, interaction and error sums of squares are 256, 96, 64 and 60. Their df are 2, 1, 2 and 18. Assume independent common-variance Normal errors. Use  $t_{.975,18} = 2.100922040$  and the Scheffé multiplier  $\sqrt{2F_{.95;2,18}} = 2.666292237$ .

1. Reconstruct the full ANOVA table, including total SS and df, all mean squares and the three F ratios. The interaction p-value is .0015. State and interpret its joint null.
2. The schedule marginal means are 8 and 12, but the conditional schedule differences are 0, 4 and 8. Explain what the practice main effect averages and why it does not describe every condition.
3. Form the pointwise interval for Answer's 4-point schedule difference using  $SE = \sqrt{5/3}$ . State the family limitation of this single-comparison interval.
4. After inspecting the plot, a reader selects Answer's schedule difference minus Rereading's, estimated as 4. Calculate its SE from four independent cell means of size 4 and  $MS_E = 10/3$ . Form its Scheffé interval and identify the protected family.

5. Explain why the interval in part 4 can contain zero when the omnibus interaction test is significant, and why that interaction-only family does not also protect part 3's simple effect.

### Exercise 7: Repair a claim about teaching (Integrative)

Repair this claim about the invented study:

Spaced practice improves every student's learning by four points, whatever feedback is provided. The significant interaction proves that Answer feedback benefits more from spacing than Rereading, and the results apply to all students.

All 24 quiz scores are observed. The stipulated design allocates four different students to each cell. The equal-weight practice main effect is 4.0000 points; conditional schedule differences are 0.0000, 4.0000 and 8.0000. The interaction test gives  $F(2, 18) = 9.6000$ ,  $p = .0015$ . The selected Answer-minus-Rereading difference of schedule differences is 4.0000, with a 95% interaction-family Scheffé interval  $[-0.8680, 8.8680]$ .

1. Identify at least four words or claims that exceed the supplied statistical and design information.
2. Write a bounded replacement retaining the conditional pattern, the joint interaction result, the selected estimate and interval, the 0–20 quiz scale, and the invented nature of the example.
3. Explain what random allocation could support in a real study and what it would not establish about population sampling or individual treatment effects.
4. Explain how class-level assignment, repeated observations or missing outcomes would affect the information needed for the analysis. Do not invent losses in the present dataset. State why four observations per cell limit diagnostic reassurance.

### Exercise 8: Match the analysis to the design (Integrative)

The Massed Rereading scores are 4, 5, 7, 8 and the Massed Answer scores are 6, 7, 9, 10. They belong to eight different students. The pooled within-group variance is  $10/3$ . For these matched classical calculations, the two-sided p-value is .1723.

1. Calculate Rereading minus Answer, its SE, the pooled t statistic and the corresponding one-way F statistic. State both tests' df and explain why their p-values agree although F loses direction.
2. Suppose instead that two measurements came from each of four students. Explain why a paired analysis would use within-student differences and why the independent-group calculation in part 1 would no longer be the appropriate starting point. Name the dependence information needed to calculate its SE.

3. Explain how assigning one class to each condition within each school creates a blocked comparison. Identify the comparison units in Chapter 3's paired-school illustration and distinguish an allocation block from a sampling stratum.
4. A researcher proposes measuring a fractions pretest before assigning feedback. Explain what baseline adjustment adds, why an adjusted comparison needs a specified common value or averaging distribution, and why adjustment alone does not establish a causal interpretation.

## Appendix A: Reproducing the ANOVA Arithmetic

The calculations here use the same scores as the main text. They show why the decompositions work and how the two-group identity follows. The numerical identities describe observed data; the model conditions are added when a reference distribution is used.

### A.1 Reconstruct the one-way decomposition

For the Massed slice, the grand mean is 8. The table separates each group's contribution to the total.

**Table 16**

*Contributions to the one-way sums of squares.*

| Feedback             | Mean    | $n_j(\bar{y}_j - 8)^2$ | Within-group<br>SS | Total<br>deviations<br>from 8 |
|----------------------|---------|------------------------|--------------------|-------------------------------|
| Rereading            | 6.0000  | 16.0000                | 10.0000            | 26.0000                       |
| Answer feedback      | 8.0000  | 0.0000                 | 10.0000            | 10.0000                       |
| Explanatory feedback | 10.0000 | 16.0000                | 10.0000            | 26.0000                       |
| Total                |         | 32.0000                | 30.0000            | 62.0000                       |

For Rereading, the within-group calculation is

$$(4 - 6)^2 + (5 - 6)^2 + (7 - 6)^2 + (8 - 6)^2 = 4 + 1 + 1 + 4 = 10.$$

Its total squared deviation from the grand mean is

$$(4 - 8)^2 + (5 - 8)^2 + (7 - 8)^2 + (8 - 8)^2 = 16 + 9 + 1 + 0 = 26.$$

The between-group contribution is  $4(6 - 8)^2 = 16$ , and  $26 = 16 + 10$ . The same calculation for Explanatory feedback gives 26 because its group mean is equally far above 8. The Answer mean equals the grand mean, so its between-group contribution is zero although its individual scores still vary.

The general proof begins by squaring the deviation split:

$$\begin{aligned}(y_{ij} - \bar{y})^2 &= (\bar{y}_j - \bar{y})^2 + (y_{ij} - \bar{y}_j)^2 \\ &\quad + 2(\bar{y}_j - \bar{y})(y_{ij} - \bar{y}_j).\end{aligned}$$

Within group  $j$ , its mean departure is constant. Summing the cross-products therefore gives

$$2(\bar{y}_j - \bar{y}) \sum_i (y_{ij} - \bar{y}_j) = 0.$$

The last sum is zero by the definition of a group mean. Summing the remaining terms over groups establishes  $SS_T = SS_B + SS_W$ . For the individual score 4, however, the squared total deviation is 16 while the two squared components sum to  $4 + 4 = 8$ ; its cross-product contributes another 8. The cancellation is a property of the full sum, not of each record separately.

Dividing by  $N - 1$  yields the total sample variance  $62/11 = 5.6364$ . By contrast, the common within-group variance estimate is  $30/9 = 3.3333$ . These denominators and targets differ: the first describes variation around one grand mean, the second estimates error variation around the three fitted group means.

## A.2 Reconstruct the balanced two-way table

The six cell means are (6, 6), (8, 12) and (10, 18) in feedback rows and practice columns. For each observed score,

$$\begin{aligned}y_{ijk} - \bar{y} &= (\bar{y}_{j\cdot} - \bar{y}) + (\bar{y}_{\cdot k} - \bar{y}) \\ &\quad + (\bar{y}_{jk} - \bar{y}_{j\cdot} - \bar{y}_{\cdot k} + \bar{y}) \\ &\quad + (y_{ijk} - \bar{y}_{jk}).\end{aligned}$$

The four parts are the feedback departure, practice departure, interaction departure and individual residual. Consider the Explanatory-Spaced student scoring 16. The grand mean is 10, the two marginal means are 14 and 12, and the cell mean is 18:

$$16 - 10 = (14 - 10) + (12 - 10) + (18 - 14 - 12 + 10) + (16 - 18) = 4 + 2 + 2 - 2.$$

The observed total departure is 6. Balance and the zero-sum constraints make the cross-products cancel when the full set of squared decompositions is summed. This yields the four components in the ANOVA table.

**Table 17**

*Add the interaction departures to reconstruct every cell mean.*

| Feedback             | Additive<br>Massed | Additive<br>Spaced | Interaction<br>departures | Observed<br>means |
|----------------------|--------------------|--------------------|---------------------------|-------------------|
| Rereading            | 4                  | 8                  | 2, -2                     | 6, 6              |
| Answer feedback      | 8                  | 12                 | 0, 0                      | 8, 12             |
| Explanatory feedback | 12                 | 16                 | -2, 2                     | 10, 18            |

The feedback term uses two free marginal departures, the practice term one, and the interaction two. The full model therefore fits one grand mean plus  $2 + 1 + 2 = 5$  effects, equivalent to six separate cell means. Within each of six cells, the four residuals have one sum constraint, leaving  $6(4 - 1) = 18$  error df.

The total can also be checked directly by centering every score at 10. The six cells contribute 74, 74, 26, 26, 10, 266 in table order, which sum to 476. Equivalently, their mean departures from 10 are -4, -4, -2, 2, 0, 8, so

$$SS_T = 4(16 + 16 + 4 + 4 + 0 + 64) + 60 = 416 + 60 = 476.$$

The between-cell quantity 416 divides into feedback 256, practice 96 and interaction 64. This gives the independent checks  $256 + 96 + 64 + 60 = 476$  and  $2 + 1 + 2 + 18 = 23$ .

### A.3 Derive the matched pooled-t identity

For two independent groups, let  $\widehat{\Delta\mu} = \bar{y}_1 - \bar{y}_2$  and  $N = n_1 + n_2$ . Their grand mean is

$$\bar{y} = \frac{n_1\bar{y}_1 + n_2\bar{y}_2}{N}.$$

Subtracting it from each group mean gives

$$\bar{y}_1 - \bar{y} = \frac{n_2}{N}\widehat{\Delta\mu}, \quad \bar{y}_2 - \bar{y} = -\frac{n_1}{N}\widehat{\Delta\mu}.$$

Consequently,

$$\begin{aligned}
SS_B &= n_1 \left( \frac{n_2}{N} \widehat{\Delta\mu} \right)^2 + n_2 \left( -\frac{n_1}{N} \widehat{\Delta\mu} \right)^2 \\
&= \frac{n_1 n_2}{N} \widehat{\Delta\mu}^2.
\end{aligned}$$

There is one numerator df, so  $MS_B = SS_B$ . Define the pooled variance  $s_p^2 = SS_W/(N - 2)$ . Then

$$F = \frac{SS_B}{s_p^2} = \frac{\widehat{\Delta\mu}^2}{s_p^2(1/n_1 + 1/n_2)} = t^2.$$

For the Rereading/Answer subset,  $n_1 = n_2 = 4$ ,  $\widehat{\Delta\mu} = -2$ ,  $SS_B = 8$ ,  $SS_W = 20$  and  $s_p^2 = 20/6 = 10/3$ . Thus  $F = 8/(10/3) = 2.4000$ , equal to the square of  $t = -2/\sqrt{5/3} = -1.5492$ . Under the matched Normal common-variance model, the upper-tail  $F(1, 6)$  probability equals the two-sided  $t_6$  probability, .1723.

This algebra does not supply a paired-data SE: paired inference works on within-pair differences and their variation. A Welch test also has its own variance estimate and reference df. The identity requires matching the data, pooled error estimate, degrees of freedom and two-sided alternative.

## Appendix B: Families, Weighting and Magnitude

### B.1 What each interval family covers

A coverage statement needs both a target and a family. The following procedures used in the chapter make different promises.

**Table 18***The family is part of the inferential statement.*

| Procedure            | Family specified here                                            | Interpretation                                                                          |
|----------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Pointwise t interval | One comparison specified without selection over relevant results | Covers that target at the stated rate under the model                                   |
| Bonferroni           | The two declared                                                 | Joint coverage of both targets is at least .95 under the model                          |
| Tukey                | Rereading-referenced comparisons                                 | Simultaneous coverage for the all-pairs family                                          |
| Scheffé              | All three pairwise feedback differences                          | Simultaneous coverage across this entire space permits a selected interaction direction |

For the selected difference of differences, retain the same estimate 4.0000,  $SE = \sqrt{10/3}$  and 18 error df throughout. Only the calibration changes:

$$\begin{aligned} \text{Pointwise: } 4 \pm 2.100922040\sqrt{10/3} &\approx [0.1643, 7.8357], \\ \text{Interaction family: } 4 \pm 2.666292237\sqrt{10/3} &\approx [-0.8680, 8.8680]. \end{aligned}$$

The pointwise interval excludes zero and the simultaneous interval does not. Selecting the narrower one because it produces a preferred conclusion would discard the selection protection. The width difference reflects the broader coverage obligation, not a change in the observed contrast.

The interaction-only family has two dimensions here. It is narrower than all possible contrasts among six arbitrary cell means, which would have five dimensions. An interaction contrast cancels both main-effect components through its row and column weight sums. The difference of differences does this; a single schedule effect within one feedback condition does not. This explains why one multiplier cannot be applied indiscriminately to every comparison.

The two one-way reference estimates also illustrate dependence. Writing them as  $\bar{Y}_A - \bar{Y}_R$  and  $\bar{Y}_E - \bar{Y}_R$ , their covariance under independent group means is  $\text{Var}(\bar{Y}_R) = \sigma^2/4$ . A low reference mean increases both differences. Bonferroni remains valid because its joint-coverage bound uses the sum of individual failure probabilities without assuming independence.

## B.2 Unequal-cell weights and effect-size denominators

In the chapter's balanced data, equal-cell and student-weighted marginal means coincide. For unequal cell counts, define them separately:

$$\bar{y}_{j\cdot}^{\text{students}} = \frac{\sum_k n_{jk} \bar{y}_{jk}}{\sum_k n_{jk}}, \quad \bar{y}_{j\cdot}^{\text{equal cells}} = \frac{1}{K} \sum_k \bar{y}_{jk}.$$

The first describes the mixture of students observed in that feedback group. The second gives each of the  $K$  practice conditions the same weight. Neither is automatically the right target for every question.

For an illustration, retain Answer's cell means 8 and 12 but imagine counts of 2 Massed and 6 Spaced students. The equally weighted mean is  $(8 + 12)/2 = 10.0000$ ; the student-weighted mean is  $(2 \times 8 + 6 \times 12)/8 = 11.0000$ . These are hypothetical weights, not changes to the actual four-per-cell dataset. A comparison of feedback groups using different schedule mixtures can combine a feedback comparison with a weighting difference. Adjusted means in later chapters make their common profile or averaging distribution explicit for this reason.

The main one-way effect-size summary was  $\eta^2 = SS_B/SS_T = .5161$ . A common reference correction, **omega squared**, subtracts an estimated noise contribution from the between-group sum of squares:

$$\omega^2 = \frac{SS_B - (J - 1)MS_W}{SS_T + MS_W} = \frac{32 - 2(10/3)}{62 + 10/3} = .3878.$$

The subtraction recognizes that group means vary even under equal model means. This correction is motivated by the common-variance model; it is not the same descriptive fraction as eta squared, and it does not establish causation. In some samples its numerator can be negative, illustrating its role as an estimator rather than a literal nonnegative share of observed variation. The raw mean differences remain necessary for understanding the outcome scale.

In factorial software output, **partial eta squared** often uses a term-specific denominator:

$$\eta_{\text{partial, term}}^2 = \frac{SS_{\text{term}}}{SS_{\text{term}} + SS_E}, \quad \eta_{\text{term}}^2 = \frac{SS_{\text{term}}}{SS_T}.$$

They agree in one-way ANOVA because the total contains only that factor and error. In our factorial model, interaction has  $64/(64 + 60) = .5161$  partial eta squared but  $64/476 = .1345$  ordinary eta squared. The former excludes the other factor components from its denominator. Always name the summary and its denominator before interpreting a software effect-size column.

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